

Table 1 – Comparison of Temperature Transducers

Type	Cost	Output	Linearity	Temp. Range (°C)	Individual Sensor Range (K)	Readily Achievable Accuracy (K)
TC	medium	low	reasonable	-250 to 2500	≈1000	±1.0
PRT	high	low	good	-200 to 850	≈500	±0.1
Thermistor	low	high	poor	-50 to 250	≈100	±1.0

3.3 Introduction to Rotational Speed Measurement Devices

The measurement of rotational speed, while not as commonplace as pressure or temperature measurements, is still widespread. It has application to rotating machinery such as internal combustion engines, turbomachinery (including wind turbines and jet engines) and aircraft attitude and motion.

For machines that operate only in 1-degree-of-rotational-freedom the speed is usually stated in units of **revolutions per minute (rpm)** rather than in Hz. In comparison, for aircraft the rotational motion occurs in 3-degrees-of-freedom (3-DoF) and is typically measured in radians per second (rad/s) by means of a **gyroscope**. In the interest of brevity, we will constrain this discussion to just measurements of 1-DoF rotation. A discussion of multi-DoF measurements using gyroscopes is left for future courses and study.

Today, most speed measuring systems use **pulse counting methods**. In these methods either an inductive or optical probe senses the passage of a feature as it rotates past and produces a corresponding train of pulses in time. The feature being detected depends on the specific design and application but could be the teeth on a gear wheel (or even the flats of a hexagonal nut), or an etched or anodized pattern on a shaft (for optical detectors). Pulse counting can either be digital (accurate) or analogue (less accurate) as follows –

1. Digital Pulse Counting: in these methods the time between pulses is recorded using a **quartz crystal oscillator** which typically have an accurate to better than ±0.02 %. The pulse time is then scaled to rpm digitally.
2. Analogue Pulse Counting: in these methods a dedicated integrated circuit is used which averages the pulses and converts the pulse frequency to a voltage.

An example such device is the *Texas Instruments LM2917-N*. The voltage output